

Developing Embodied Conversational Agents in the Unreal Engine: The FANTASIA Plugin

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ABSTRACT

The fast-evolving industry of games pushes the technology behind audio, graphics and controllers to evolve at a very high speed. The front-end of interactive experiences such as games sets standards that must be met also by research teams working on Artificial Intelligence. It is, therefore, important, to devise strategies to let researchers concentrate on interaction *models* while being up to date with interface design standards. FANTASIA is designed to extend the functionalities offered by a high-profile industrial game engine, the Unreal Engine, to obtain an advanced development environment to develop Embodied Conversational Agents. The FANTASIA plugin is open source and is freely available on GitHub¹ together with tutorial series on a dedicated Youtube channel.

CCS CONCEPTS

• **Human-centered computing** → **User interface toolkits; User interface management systems.**

¹<https://github.com/antori82/FANTASIA>

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KEYWORDS

Embodied Conversational Agents, Unreal Engine, Artificial Intelligence

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1 INTRODUCTION

In the last few years, industry-grade game engines such as Unity 3D² and the Unreal Engine 4³ (UE4) have been made available to the general public. Differently from previous experiences, which required strong programming skills to actually make use of the engines, the newest versions of these tools pay extreme attention to entry-level users, developing user-friendly interfaces for developers to access their most powerful technologies. As a matter of fact, the interest of game engine developers is now to make features such as environment navigation, animation logic, etc. . . more accessible and easy to integrate into complex workflows. While the industrial effort concentrates on interface quality, development speed and technological integration, researchers in the Artificial Intelligence (AI) field are mainly interested in the models underlying the interactive processes. These aspects are less explored by the industry,

²<https://unity3d.com/>

³www.unrealengine.com

making the main activity of AI and Human-Machine Interaction (HMI) researchers complementary to the industry activity. Integration between the two efforts has the potential to bring benefits to both realities. Specifically, the practical advantages of designing an approach centred on freely available, continuously evolving, technology lie in the possibility, for researchers, of concentrating on interaction management models, rather than on technicalities brought in either by the necessity to compete with the design quality standards set by the entertainment industry or by the need to integrate a new control device. The new version of the FANTASIA plugin [2] concentrates on supporting the development of interactive experiences that are based on natural interaction and have an impact on social processes. In this view, a framework to integrate game engines with tools that are important for studies on natural interaction is relevant, to support AI and HMI research. The foundations of this effort can be summarised as follows:

- Build upon the interest of the gaming industries in providing advanced environments to develop interactive experiences.
- Let researchers concentrate on Artificial Intelligence models while avoiding being left behind with respect to industrial standards set on design.

In this demo, we summarise the features offered by FANTASIA and we present an example of the development experience of conversational agents using the presented functionalities.

2 FANTASIA FEATURES

In this Section, we summarise the main features made available by FANTASIA to support the development of Embodied Conversational Agents. All the presented functionalities are integrated with the Blueprint visual scripting language to provide a coherent development experience inside the game engine.

2.1 Natural Language Processing

FANTASIA provides a set of nodes to integrate the Unreal Engine with Azure functionalities. Specifically, both Automatic Speech Recognition (ASR) and Natural Language Understanding (NLU) are provided. Concerning NLU, FANTASIA allows microphone audio streaming from the Unreal Engine directly to the Microsoft Language Understanding Intelligent Service (LUIS).

2.2 Speech synthesis

Speech synthesis with FANTASIA is provided by offering a connection to Amazon Polly which, differently from other services, also provides viseme information, which is fundamental to implement lip-sync. Also, Amazon Polly offers vast support to the Speech Synthesis Markup Language (SSML), allowing the use of markers. When markers are used, FANTASIA raises dedicated events to indicate when marked words are being pronounced. This is important to synchronise, for example, gestures of facial expressions.

2.3 Artificial Intelligence

FANTASIA provides an integration with a graph database, Neo4j [3], and with the open source Graphical Universal Model framework (aGrUM) [1], which can be used to import, query and update Bayesian Networks. In FANTASIA, Neo4j is mainly used for

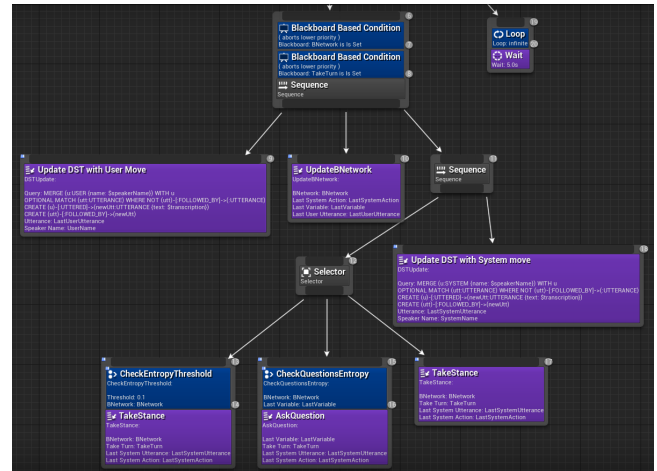


Figure 1: A UE Behaviour Tree integrated with Neo4j access and Bayesian Networks

knowledge representation while Bayesian Networks are used to implement probabilistic decision systems. Blueprint nodes to interact with Neo4j and Bayesian Networks can be used inside regular Blueprints but find their most powerful application with the UE Behaviour Trees. As both tree decorators and tasks are implemented using Blueprints, accessing modelled knowledge and a probabilistic infrastructure from inside Behaviour Trees combines the efficiency power of the Unreal AI architecture with the advanced functionalities provided by graph databases and Bayesian Networks.

3 EXAMPLE APPLICATION

The application presented to illustrate how FANTASIA supports the development of ECAs in the Unreal Engine makes use of a Metahuman character⁴ called Jason that engages users in simple conversations concerning the movies domain. Jason will be able to answer questions and support users in finding movies they may like. The example application will be developed in English and it will use voice input. The application will show how speech input can be captured and interpreted with LUIS, how the extracted information can be used to query Neo4j for the relevant data and update a Bayesian Network and how synthesised speech can be dynamically altered using the UE audio engine effects. The entire AI logic will be implemented in combination with the UE subsystem for Artificial Intelligence using Behaviour Trees, as shown in Figure 1.

REFERENCES

- [1] Gaspard Ducamp, Christophe Gonzales, and Pierre-Henri Wuillemin. 2020. aGrUM/pyAgrum: a toolbox to build models and algorithms for Probabilistic Graphical Models in Python. In *International Conference on Probabilistic Graphical Models*. PMLR.
- [2] Antonio Origlia, Francesco Cutugno, Antonio Rodà, Piero Cosi, and Claudio Zmarich. 2019. FANTASIA: a framework for advanced natural tools and applications in social, interactive approaches. *Multimedia Tools and Applications* 78, 10 (2019), 13613–13648.
- [3] Jim Webber. 2012. A programmatic introduction to Neo4j. In *Proceedings of the 3rd annual conference on Systems, programming, and applications: software for humanity*. ACM, 217–218.

⁴<https://metahuman.unrealengine.com/>